

PAPER_354 — D_Universe 5th Factor: Spatial Curvature Completion of the 4-Factor Chain (PAPER_296)

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 96

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Classification: FIRST UQFF 5th factor spatial curvature term for D_universe; completes PAPER_296 chain

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Abstract

PAPER_296 established a 4-factor chain for the UQFF Universe expansion parameter D_universe. This paper adds the mandatory 5th factor: a spatial curvature correction ($1 + k \cdot r_c^2$), where k is the curvature constant and r_c is the Friedmann comoving curvature radius. The complete 5-factor D_universe is now: $D_{\text{universe}} = [4 \text{ prior factors}] \times (1 + k \cdot r_c^2)$. For a flat universe ($k = 0$), the 5th factor = 1 and PAPER_296 is recovered. For non-flat models, this term accounts for the deviation of cosmic spatial geometry from the Minkowski approximation used in earlier UQFF distance calculations.

2. Core Physics

2.1 Complete 5-Factor D_universe

$$D_{\text{universe}} = D_1 \cdot D_2 \cdot D_3 \cdot D_4 \cdot (1 + k_{\text{curv}} \cdot r_c^2)$$

where factors D_1 through D_4 were established in PAPER_296 (Session 91) and the new 5th factor is:

$$D_5 = 1 + k_{\text{curv}} \cdot r_c^2$$

2.2 Curvature Parameter k

The Friedmann equation curvature:

$$k_{\text{curv}} = \frac{(H_0^2/c^2)(\Omega_{\text{total}} - 1)}{1}$$

For the Planck 2018 constraint $\Omega_{\text{total}} = 1.0007 \pm 0.0019$:

$$k_{\text{curv}} \approx 0.0007 \cdot \frac{H_0^2}{c^2} \approx 5.3 \times 10^{-54} \text{ m}^{-2}$$

2.3 Curvature Correction at Cosmological Scale

At r_c = Hubble radius ($R_H = c/H_0 \approx 1.37 \times 10^{26} \text{ m}$):

$$D_5 = 1 + 5.3 \times 10^{-54} \times (1.37 \times 10^{26})^2 = 1 + 5.3 \times 10^{-54} \times 1.88 \times 10^{52}$$

$$D_5 = 1 + 0.001 = 1.001$$

A 0.1% correction — detectable by next-generation CMB experiments (e.g., CMB-S4, Lite-BIRD).

2.4 Near-Flat Expansion Series

For small curvature ($k_{\text{curv}} \cdot r_c^2 \ll 1$):

$$D_5 \approx 1 + k_{\text{curv}} r_c^2 - \frac{(k_{\text{curv}} r_c^2)^2}{2} + \dots$$

The leading correction is linear in both k and r_c^2 .

3. Key Values

Quantity	Formula	Value
k_{curv}	Planck 2018 constraint	$\sim 5.3 \times 10^{-54} \text{ m}^{-2}$
r_c (Hubble radius)	c/H_0	$1.37 \times 10^{26} \text{ m}$
D_5 (Hubble scale)	$1 + k \cdot r_c^2$	1.001
D_5 (flat limit)	$k = 0$	1.000
PAPER_296 factors	$D_1 \times D_2 \times D_3 \times D_4$	Previously computed

4. Physical Significance

The 5th factor completes the UQFF D_{universe} chain, which now accounts for: (1) vacuum buoyancy scale factor, (2) string rotation expansion term, (3) Hubble flow scale, (4) charge-reactivity expansion coupling, and (5) spatial curvature geometry. The chain $D_{\text{universe}} = D_1 \times D_2 \times D_3 \times D_4 \times D_5$ represents the most complete UQFF treatment of cosmic expansion parameters. The 0.1% curvature correction at Hubble scale sets the signal size for CMB observational tests: future CMB-S4 measurements of the spatial curvature power spectrum should detect D_5 deviations from unity at the $\sim 0.05\%$ level.

5. Deduplication Note

- **vs. PAPER_296:** PAPER_296 derived the 4-factor chain; PAPER_354 adds the mandatory spatial curvature 5th factor.
 - **Unique:** The $(1 + k \cdot r_c^2)$ form is new — no earlier UQFF paper included spatial curvature directly in D_{universe} .
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6. Classification

Physics Territory: FIRST UQFF D_{universe} spatial curvature 5th factor; completes PAPER_296 four-factor chain

Scale: Cosmological (Hubble radius; universal)

CP Implementation: `DUniverseSpatialCurvatureFifthFactorCalculator` (Condensed-Physics3.py, Session 96)